

Localized Agronomic Knowledge Retrieval in Nigeria

Smallholder farmers in Nigeria produce much of the nation's food supply, but they often struggle to get timely, accurate agronomic advice when they need it most. Crop diseases, pest infestations, soil degradation, and unpredictable climate conditions frequently go unaddressed, not because the knowledge doesn't exist, but because it's scattered across extension documents, research publications, and government guidelines that farmers have no easy way to search. A farmer describing symptoms in their own words — a discolored leaf, a wilting stem, dry soil — often can't find the right guidance because keyword search fails to bridge the gap between how farmers describe problems and how experts document solutions. Limited extension services, language barriers, and weak information channels compound the problem, leaving urgent agronomic issues to linger alongside routine ones.

We're building a retrieval system at the core of a Retrieval-Augmented Generation (RAG) pipeline that takes a farmer's natural language query — symptoms observed in crops, soil conditions — and returns the top-5 most relevant documents from a curated agricultural knowledge base. This solution is guided by relevance that captures real farmer language rather than exact keyword matches, localization that reflects Nigeria's specific crops (cassava, maize, rice), pests, and agro-ecological zones, and rigor in benchmarking against expert-labelled judgments rather than assumptions about what "good" retrieval looks like.

Success means a farmer typing a plain-language description of their problem finds the document that actually answers it, extension workers and farmers spend less time digging through irrelevant material, and high-relevance guidance consistently surfaces in the top results — measured directly through nDCG@5 against expert benchmarks, not just proxy accuracy.